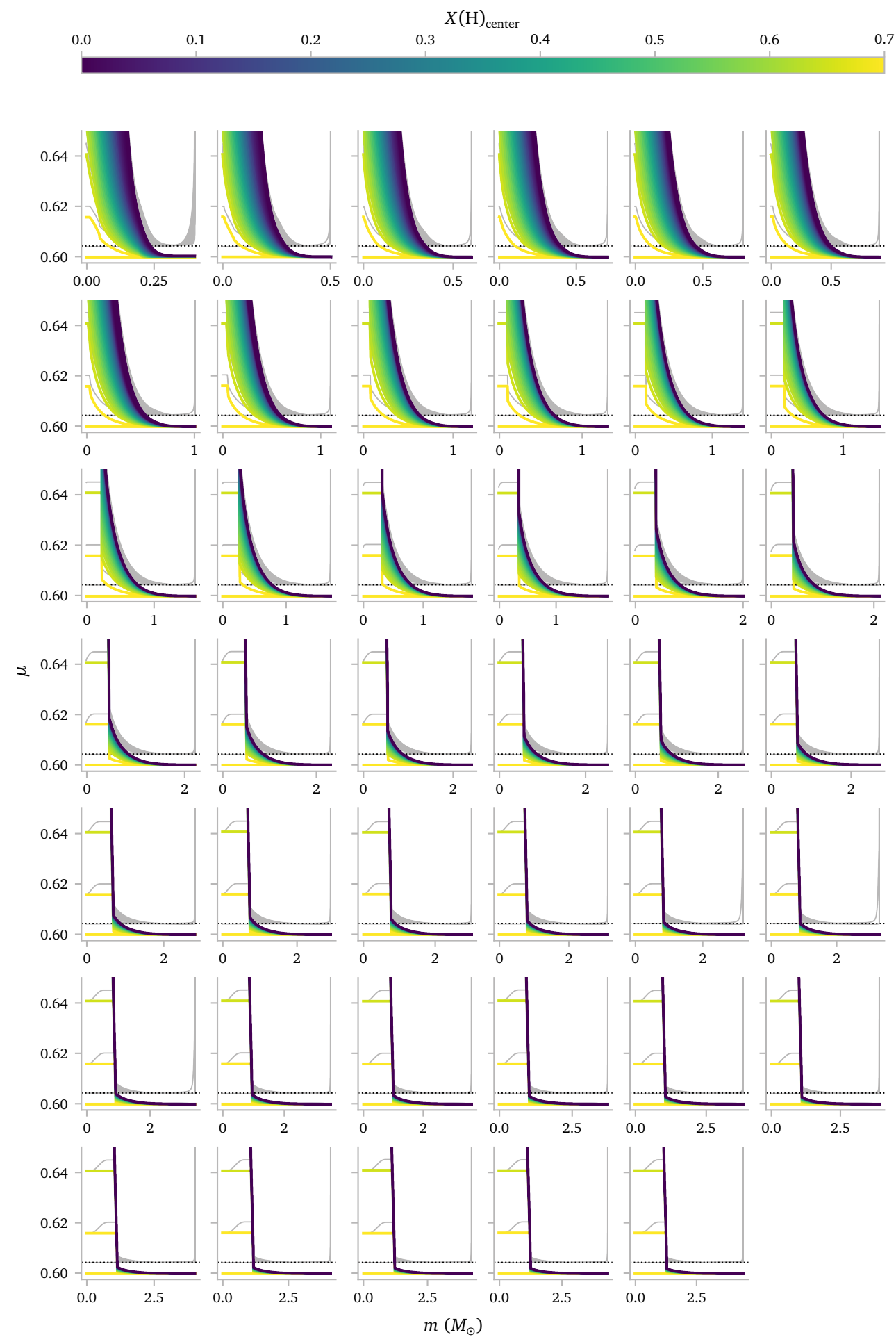


Notes Week 31

a. Fully Ionized Mean Molecular Weight

a.1 Main Sequence

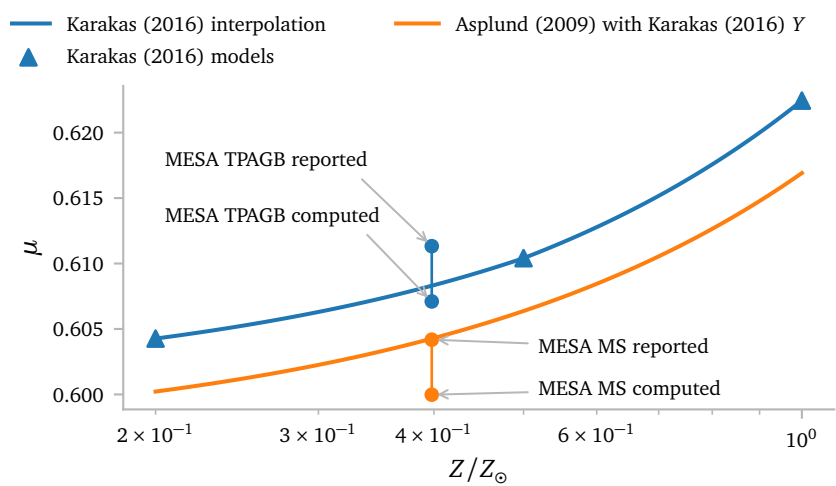
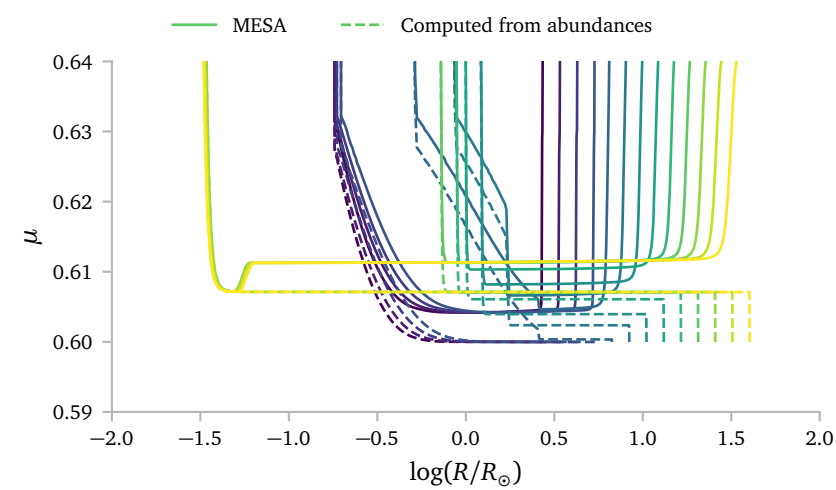
Since we are using the fully ionized mean molecular weights for the accreted material, it is only fair to compare this with the fully ionized mean molecular weights of MESA abundances.



The gray lines are the μ values as reported by *MESA*. The colored lines are the computed values from the abundances and the horizontal line is μ from [Asplund et al. \(2009\)](#). I *really* do not understand why the computed mean molecular weights from the *MESA* models are lower than reported μ from *MESA* and the computed [Asplund et al. \(2009\)](#) μ values.

I feel like the μ values reported from *MESA* are more representative because they agree with [Asplund et al. \(2009\)](#)(?)

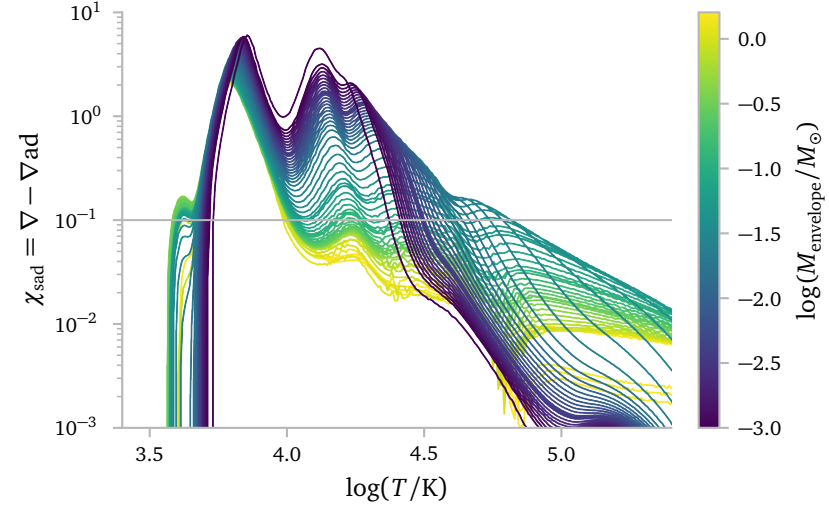
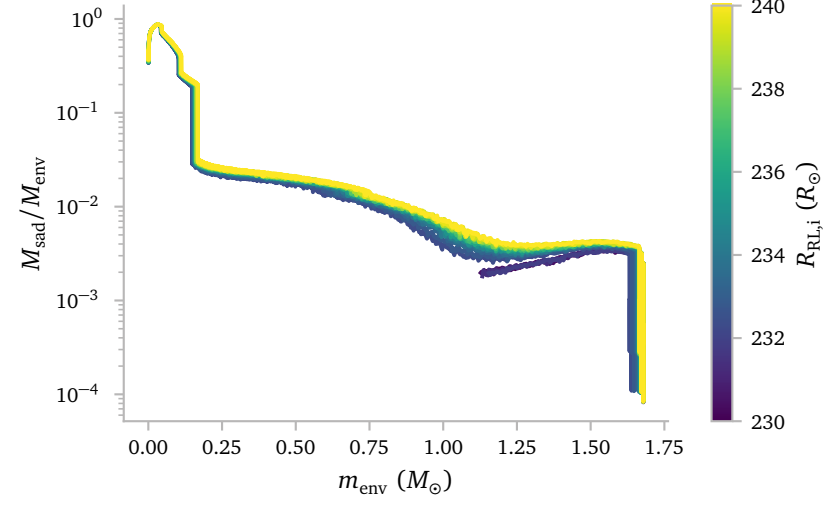
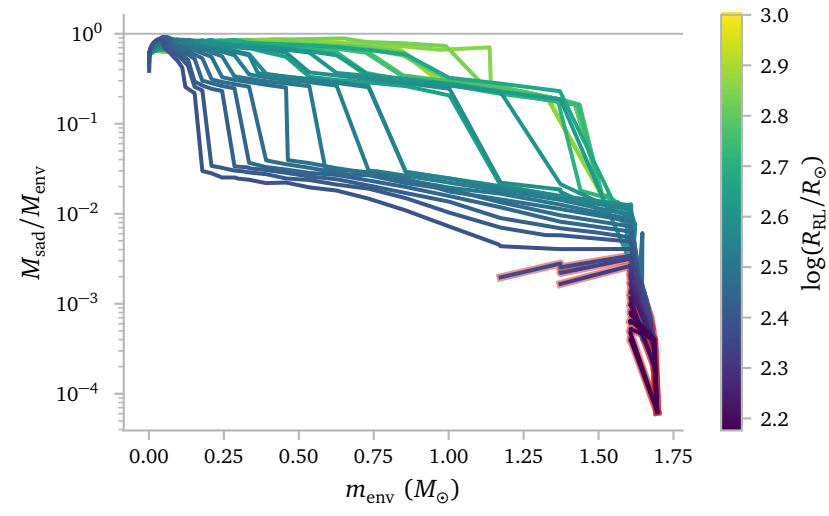
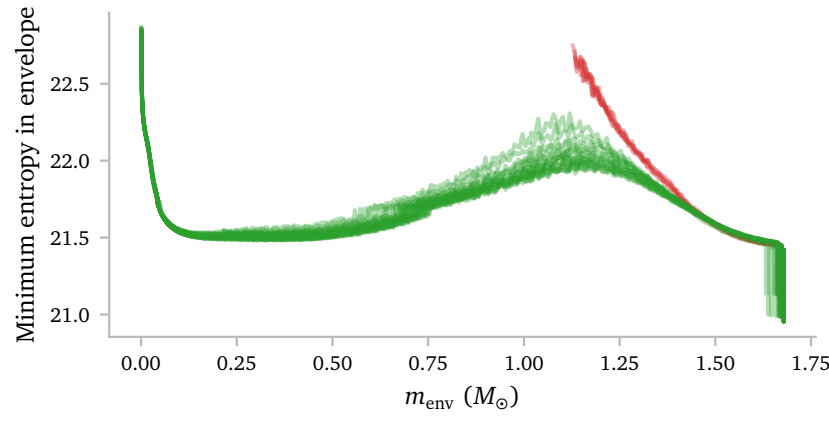
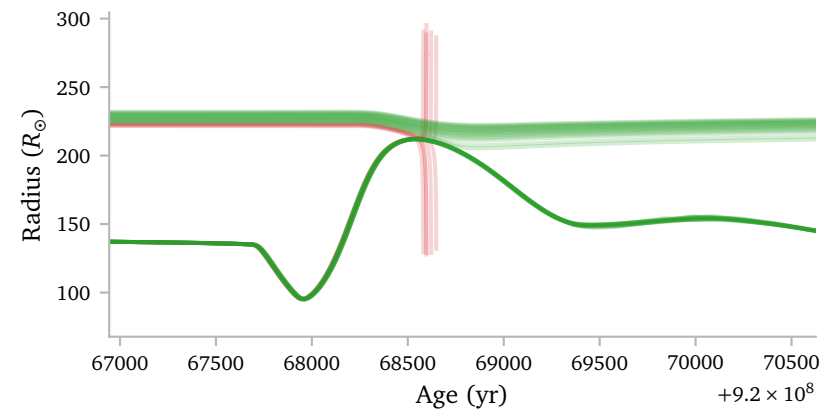
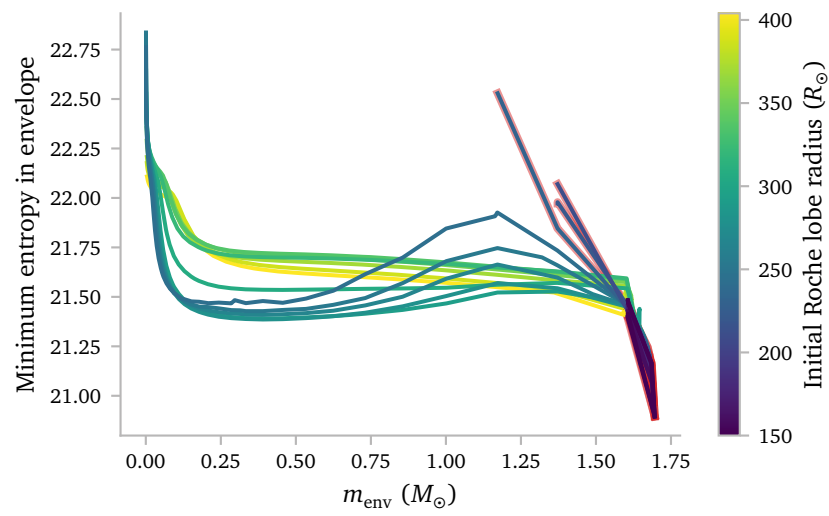
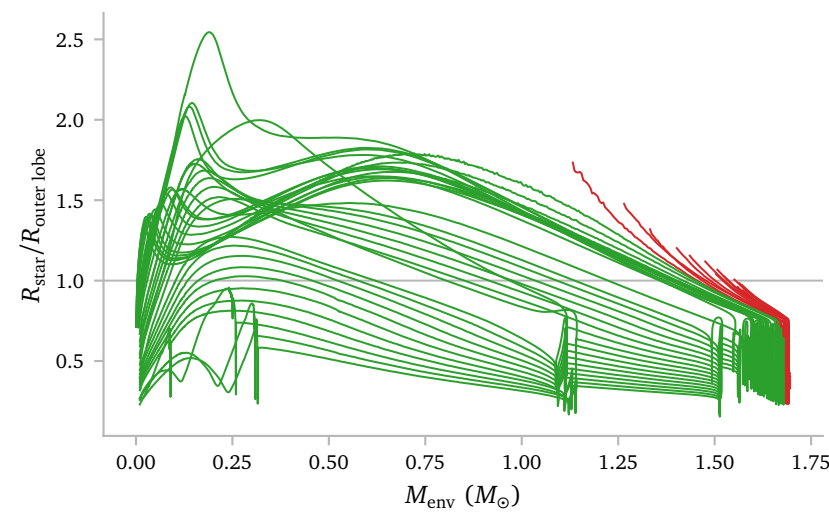
a.2 Giant branch and Thermally Pulsing Asymptotic Giant Branch



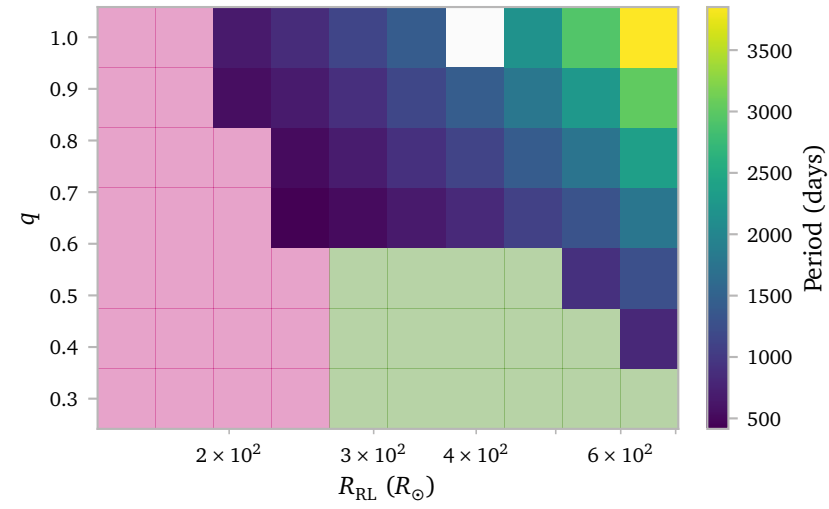
The mean molecular weights reported by the *MESA* models on the main sequence agree very well with the mean molecular weights of the [Asplund et al. \(2009\)](#).

On the TPAGB, it is exactly opposite. Here, the computed values from the abundances agrees much better with the μ computed from the [Karakas & Lugaro \(2016\)](#) abundances.

b. Failing models II

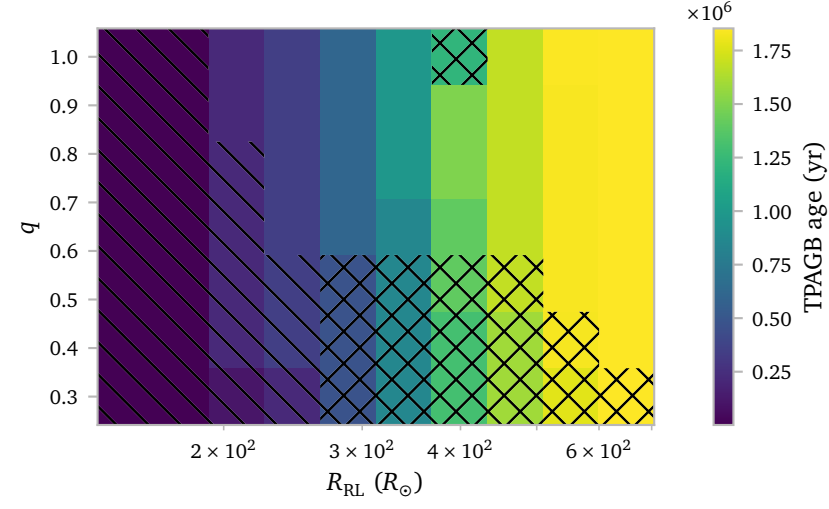


I want to return to this plot I made a while ago:

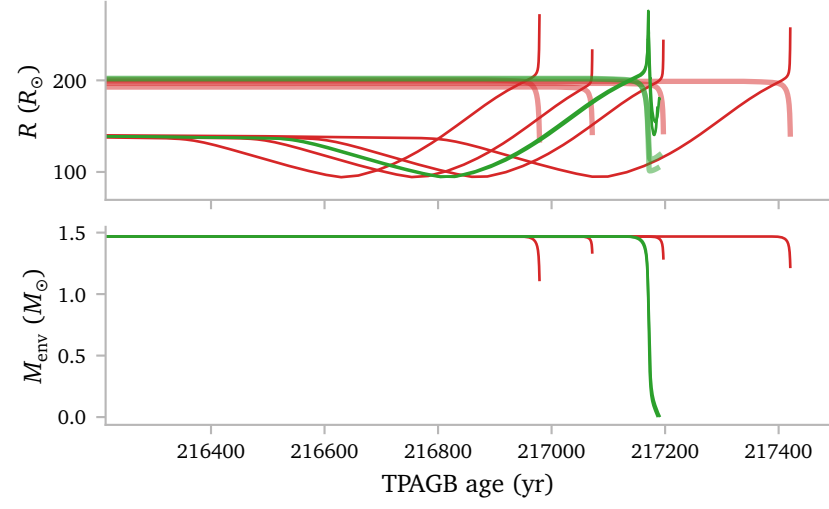


This is a grid with a $M = 2.0 M_{\odot}$ TPAGB-star, $\gamma = 1$ and $\delta = 0.5$. The pink squares are models that fail due to convergence problems. The green models end because the models spiral in.

I am curious if the models that fail, all happen in during the same thermal pulses. If however, it is possible to survive RLOF during a thermal pulse with a certain q , but not with another q .

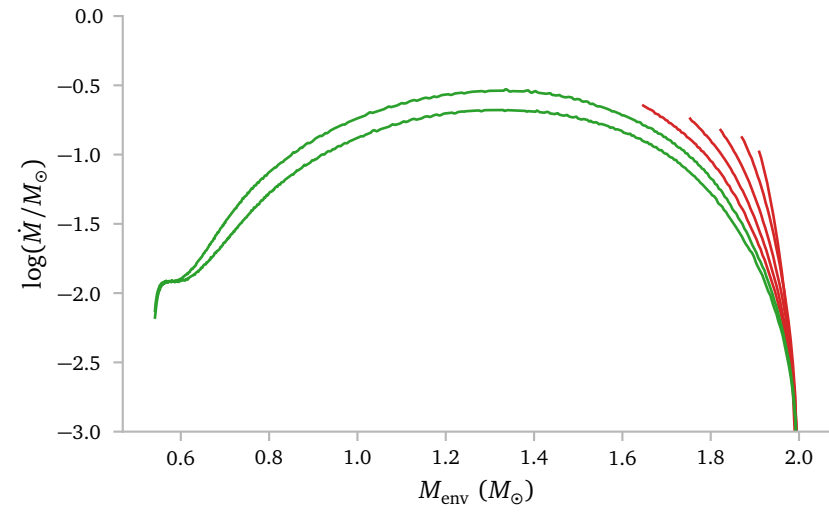


Clearly there are models that interact at roughly the same time (same color), but some of them crash, (dashed) while others do not.

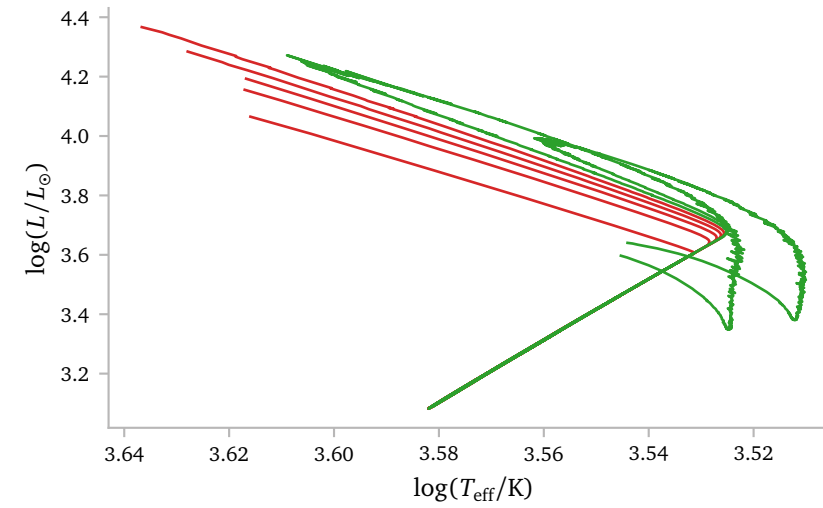


Here, I explicitly show this. The models interact during the same pulse, however, only the green lines, with the largest q , survive.

I guess the mass loss rate is another important consideration.



I show that here. The models with the largest q have the smallest $\dot{M}$ and thus strip the star slower. The differences here might seem minor, but this is a log scale so differences are actually quite large.



c. Grids

References

- Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. [ARA&A47](#), 481.
- Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. [ApJ825](#), 26.